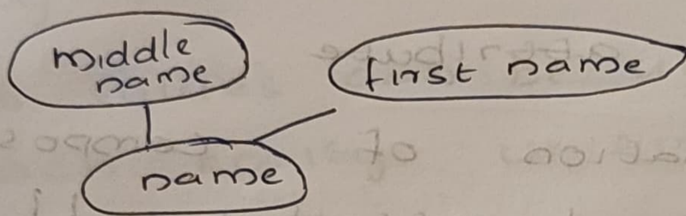
# entity - Relationship

## Types of attribute

- simple attribute :- eg: city, State, Roll no
  cannot be divided

- composite attribute
  eg:- name, address




can divided into smaller subparts

- single valued
  which having single value
  eg: Age, Gender

- multi valued Attributes
  can take more than one value
  eg: mob no, email id, address

represent email id

- Stored attribute

derivable from the birthdate attribute is called Stored

eg: date of birth

- derived attribute

Age derived is hence called derived

eg: age ; age ;

- Complex attribute ()

combination of composite &

multi valued attribute. {}

eg: Address _ phone ({phone_number},
{Address (flat no, city, state)})

- Null valued attribute

null value is a value which is not inserted but it does not hold zero value.

eg: mobil_no attribute of a person may not be having mobile phone.

- Entity type :- collection of entities that have the same attributes or properties.

- Entity set :- collection of same type of entities; ie their attributes are same is called entity set.

- key attribute.

It values can be used to identify each entity uniquely

eg:- Roll no, Age.

Relationship

Relationship type :- which is associating with different entities

Relationship set :- set of relationship of the same type

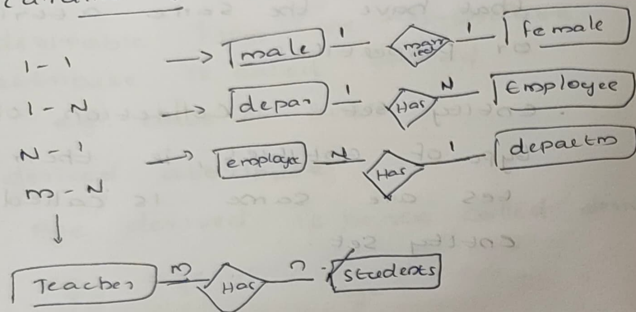
## cardinality ratio

1-1 →

1-N →

N-1 →

m-N



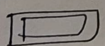

## participation constraints

Total participation

partial participation

## types of entity types

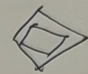
- Strong entity type → atleast one key attribute also called primary key attribute key
- weak entity type → does not have any key attribute



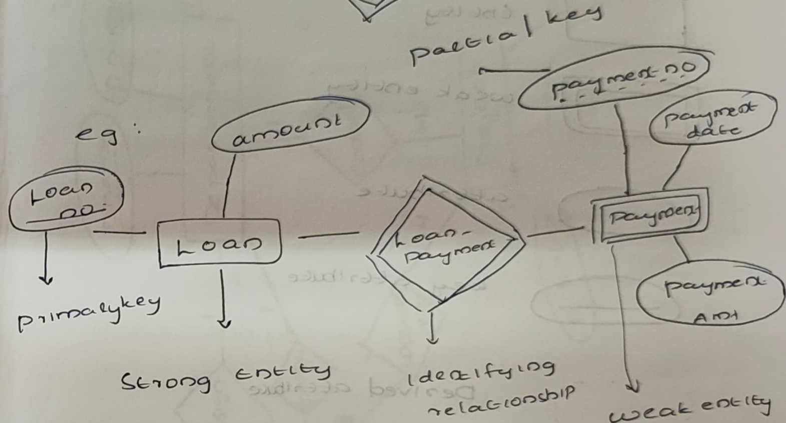
- Identifying relationship

link strong & weak entity

represent



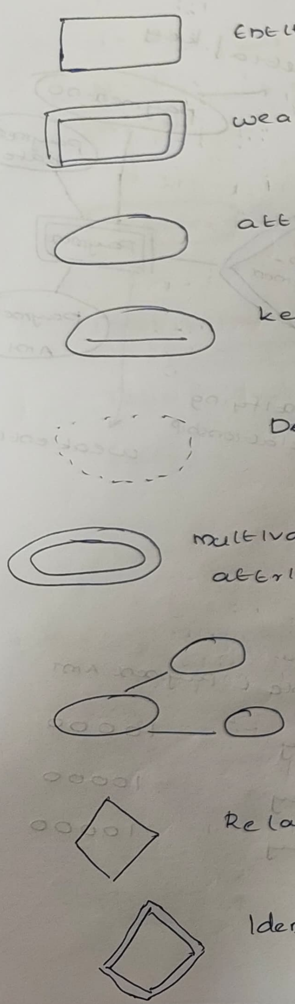
eg:




| Loan no | Amount |
|---|---|
| L1 | 100000 |
| L2 | 200000 |
| L3 | 150000 |

| payment no | payment date | payment AMT |
|---|---|---|
| P101 | 26-may | 10000 |
| P101 | 26-may | 10000 |
| P101 | 26 may | 10000 |

notations



Entity

weak entity

attribute

key attribute

Derived attribute

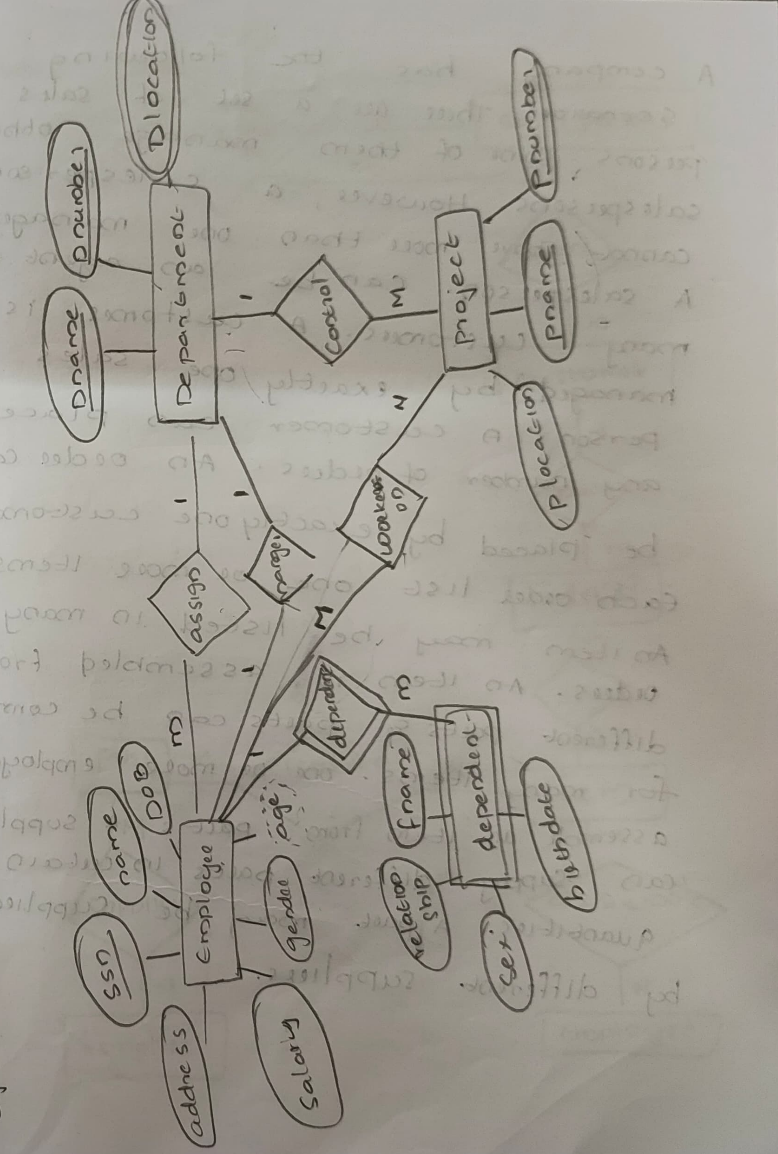
multivalued attribute

composite attribute

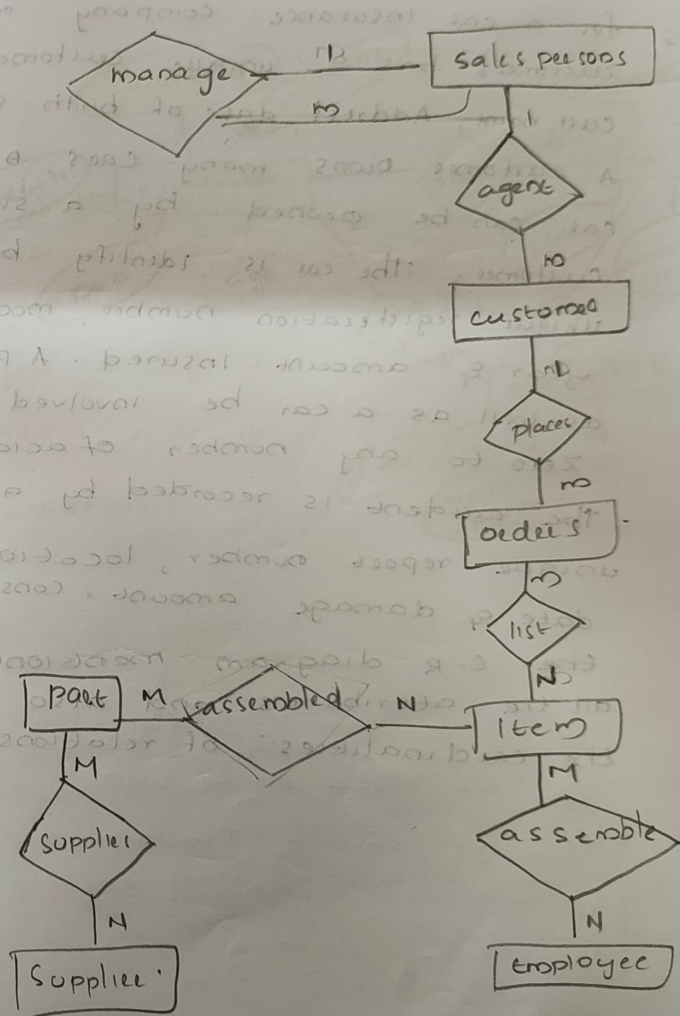
Relationship type

Identifying Relationship

eg:

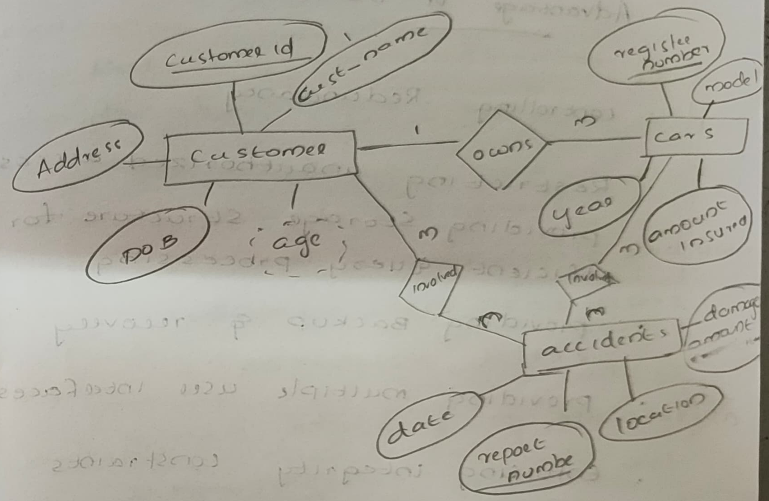


A company has the following
Scenario There are a set of sales
persons. Some of them manage other
salespersons. However, a salesperson
cannot have more than one manager
A salesperson can be an agent for
many customers. A customer is
managed by exactly one sales
person. A customer can place
any number of orders. An order can
be placed by exactly one customer
Each order list one or more items
An item may be listed in many
orders. An item is assembled from
different parts & parts can be common
for many items. one or more employee
assemble an item from parts A supplier
can supply different parts in certain
quantities. A part may be supplied
by different suppliers.



Q. for a car insurance company whose customers has a unique customer_id cust_name, Address, date of birth & age A customer owns many cars, A car can be owned by a single customer. The car is identify by unique registration number, model, year & amount insured. A person as well as a car be involved in zero to any number of acidents

· the accident is recorded by a unique report number, location, date & damage amount. Construct the E-R diagram mentioning all the attributes and also the cardinalities of relationship



## Advantage of DBMS

- controlling Redundancy
- Restricting unauthorized access
- providing Storage Structure for efficient Query processing
- providing Backup & recovery
- providing multiple user Interfaces
- Enforcing Integrity constraints

## Disadvantage of DBMS

cost of hardware & software

cost of Data conversion

cost of staff Training

Appointing Technical staff

Database Damage.

## Actors on the scene

Database administrator

Database designers

End users:

casual ~~end~~ users

. Naive or parametric end user

. sophisticated end users

. standalone users

. System Analysis & Application programmers.

The company database keep track as company employee, department and project we store Empolyee name, ssn, address, salary, gender date of birth, age. An Empolyee is assigned to one department but may work on several projects which are not necessary. Controlled by the same department. A particular Empolyee manages the department. Each department has an unique name an unique number & several locations. The department controls number of projects each of which has a unique name, unique number & a single location. we went to keep track of the dependent of each employee for insurance purpose we keep each dependents first name, sex, birth date & relation ship to employee.

Data types can be classified into 3 type

Structured → well organized

unstructed → completely unorganized

It is worst form collection of data.

Semistructed

↓

Organized but not fully

Every diffcult to retrieve the data

Datamodels

→ collection of concepts that can be used to describe the structure of database

Datamodels

→ entity

→ attribute

→ relation

conceptual data model or

highlevel data model

representation data model

or implementation data model

Low level or physical.

Query used

it hide

Hierarchial datamodel

relation datamodel

network datamodel

# Module-2.

Components of relational database

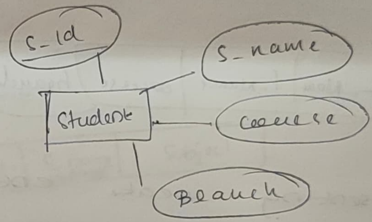
- Domain
- Tuples (rows)
- columns
- key
- Relations (Tables)

key of a relation

Each row has a value of a data item that uniquely identifies that row in the table called the key

## Converting ER diagram to relational schema.

1. Converting strong Entity set into Relational Scheme.

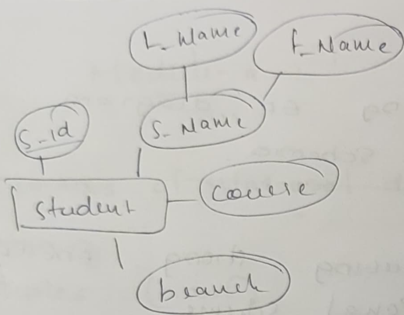



Student

| S_id | S_name | Course | branch |
|---|---|---|---|

2. Composite Attribute to Relational Scheme.

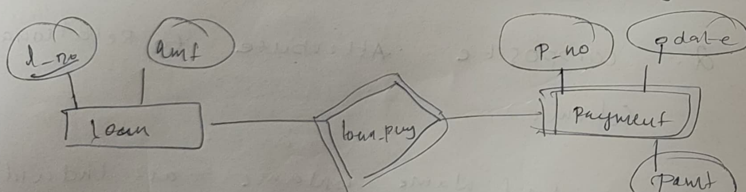
Make f_Name L_Name as Individual attribute




| | | | | |
|---|---|---|---|---|
| S_id | L_Nam | f_name | course | branch |

3. Representing weak entity set in relational scheme

- Include the primary key of strong entity set in the weak entity

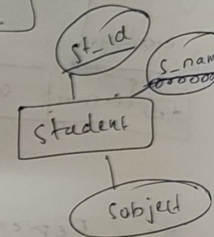



Loan

| | |
|---|---|
| l_no | amount |

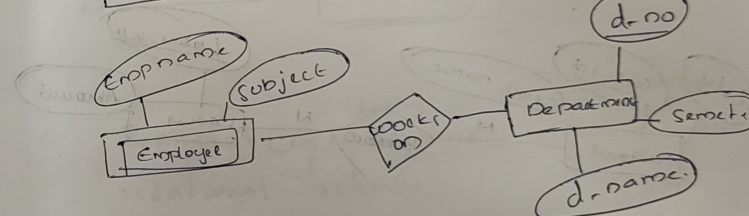
payment

| | | | |
|---|---|---|---|
| P_no | p_date | P_amt | l_no |




Student

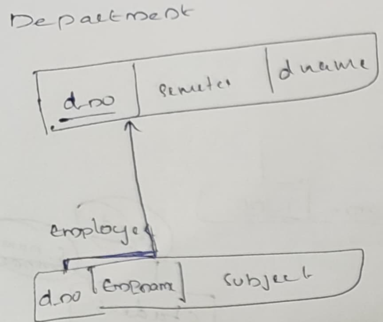
| | | |
|---|---|---|
| st_id | s_name | subject |



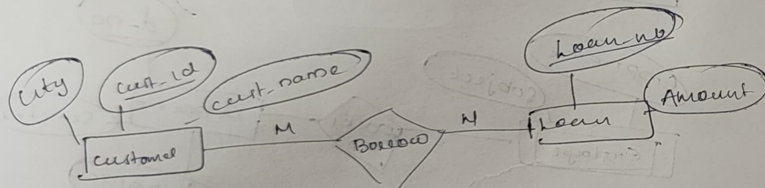

Employee

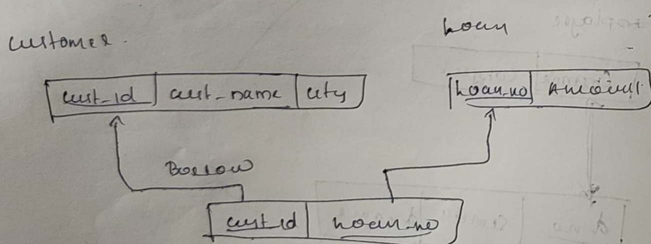
| | |
|---|---|
| Empname | subject |

| | | |
|---|---|---|
| d_no | semester | d name |

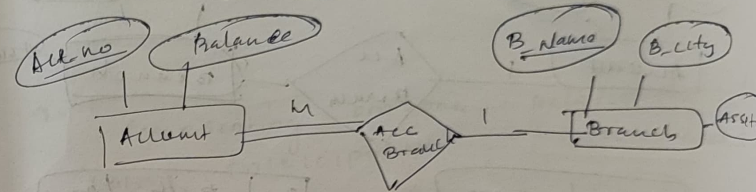



4. Representing many-many relationship in relational scheme

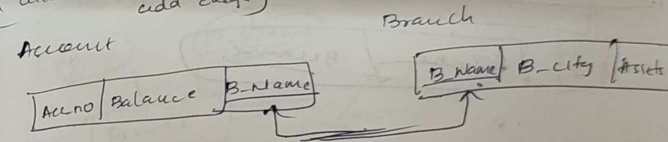


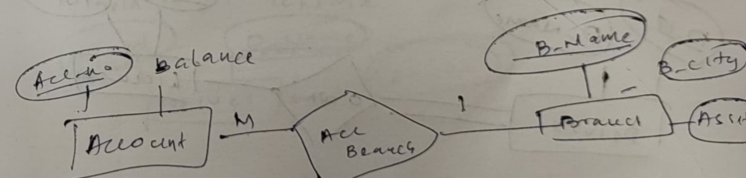

5. Representing many-one or one-many relationship with total participation in relational scheme

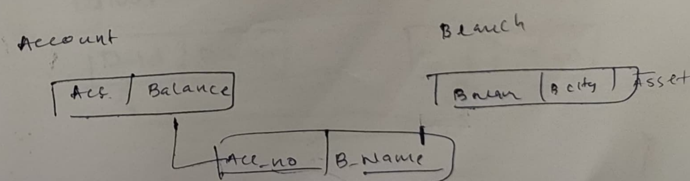



(M ulla and primary key add cheya)

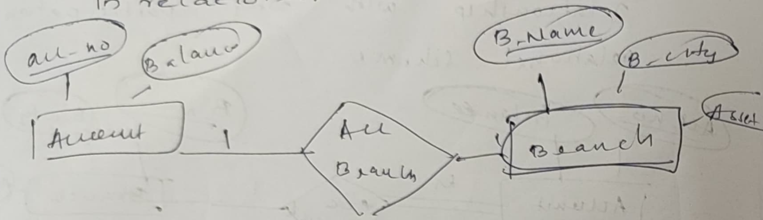


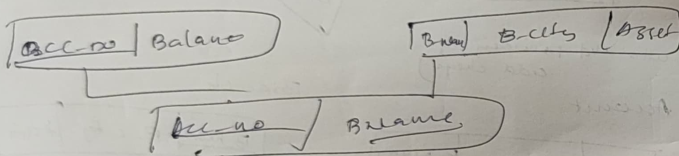

6. Representing many or one-many ~~to~~-one relationship in relational scheme.



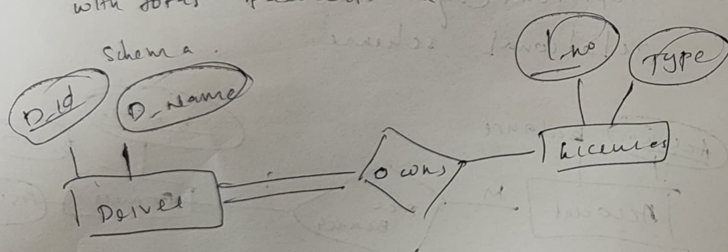



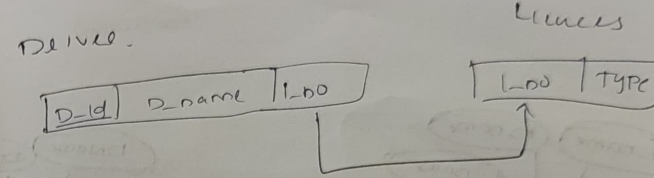
7. Representing one-one relationship in relational schema.



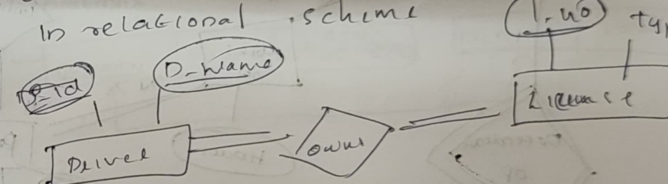



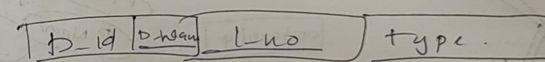

8. Representing one-one relationship with total participation in relational schema.



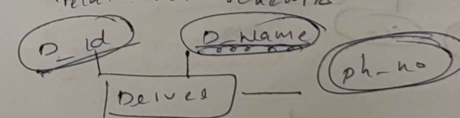



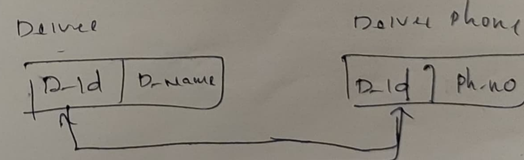

9. Representing one-one relationship with total participation at both ends in relational scheme

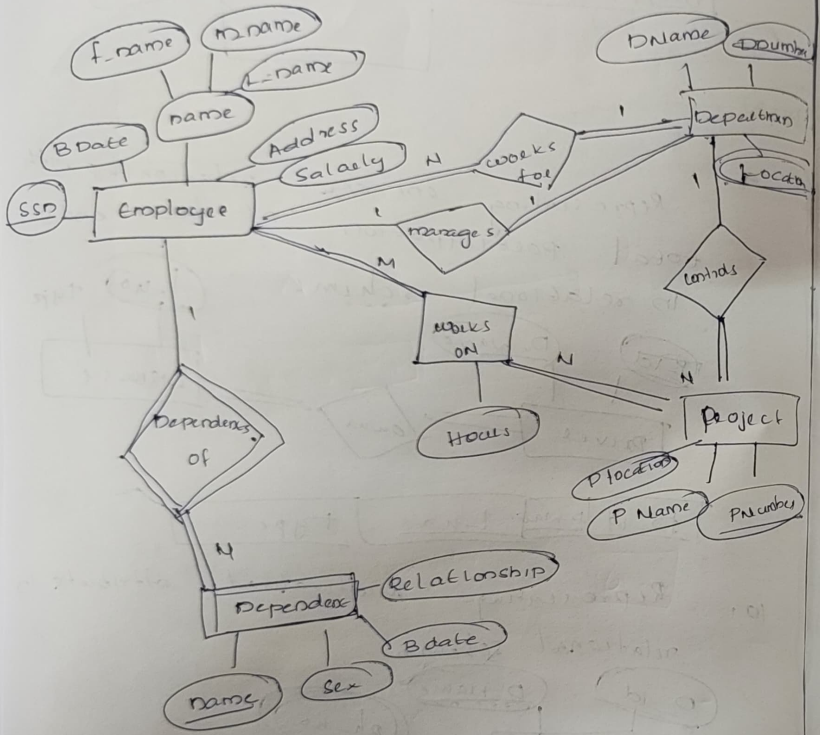



10. Representing Multivalued attribute in relational schema






Dependent

Employee

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| D_no | SSN | BDate | f_name | m name | L_name | Address | Salary |

Dependent

| | | | | |
|---|---|---|---|---|
| name | sex | Bdate | Relationship | SSN |

Department

| | | |
|---|---|---|
| D_number | DName | ~~Wo...~~ SSN |

Dept location

| | |
|---|---|
| D_number | Location |

Project

| | | |
|---|---|---|
| P_no | p name | p location |

works on

| | | |
|---|---|---|
| SSN | P_no | hours |

Stores

| | | | | |
|---|---|---|---|---|
| Store ID | Address | Manager ID | cityname | State |

Cities

| | | |
|---|---|---|
| CityName | State | HeadOffice Address |

Hold

| | | |
|---|---|---|
| StoreID | ItemID | QtyHeld |

Incities

| | | | |
|---|---|---|---|
| State | cityname | Item-ID | [illegible] |

Items

| | | | |
|---|---|---|---|
| Item-ID | Description | size | Color |

Itemordered

| | | |
|---|---|---|
| Item-ID | Or-nom | Qty ord |

Orders

| | | |
|---|---|---|
| Or-no | ord Date | Customer_name |

customer

| | | |
|---|---|---|
| customer_name | Address | Phone no |